

Secure Exit

SIMATS
ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

BAGYA VENKATESH
192324263RA

CO3-AT3-Debugging and Optimization

View Only

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QUESTIONS

Q1

ATM Withdrawal – Exception
Handling Debugging
10 marks

Q2

Hospital Registration –
NullPointerException
Debugging
10 marks

Q3

E-Commerce Product Search –
Array Exception Debugging
10 marks

Q4

Bank Account – Race Condition
Debugging
10 marks

Q5

Railway Reservation –
Synchronization Debugging
10 marks

5/5 answered

Q3

E-Commerce Product Search – Array
Exception Debugging10
marks

An online shopping application allows customers to select a product using a product number from 1 to 5.

The following program contains an array indexing error.

Task:

1. Identify the error.
2. Correct the product-number and array-index logic.
3. Handle invalid product numbers.
4. Handle non-numeric input.
5. Submit the COMPLETE corrected Java program.
6. The program must display "Search completed" after processing.

Faulty Program:

```
import java.util.Scanner;

class ProductSearch {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String[] products = {
            "Laptop",
            "Mobile",
            "Tablet",
```

```

"Headphone",
"Keyboard"
};

System.out.print("Enter product number (1-5): ");
int productNumber = sc.nextInt();

System.out.println("Selected product: "
+ products[productNumber]);

System.out.println("Search completed");
}
}

```

Correct the program so that product numbers 1, 2, 3, 4 and 5 correctly access the corresponding products.

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 import java.util.*;
2 class ProductionSearch{
3     public static void main(String[] args)
4     {
5         Scanner sc = new Scanner(System.in);
6         String[] products={"Laptop", "Mobile", "Headphone", "Keyboard"};
7         try{
8             int n=sc.nextInt();
9             if(n<1||n>5)throw new Exception("Invalid Input");
10            System.out.println("Selected product is: "+products[n-1]);
11        }
12        catch(InputMismatchException e){
13            System.out.println("Invalid Input");
14        }
15        catch(Exception e){
16            System.out.println("Invalid product number");
17        }
18        finally{
19            System.out.println("Search completed");
20        }
21    }
22 }

```

21 }



Test Input

3



Output

```
Selected product:Tablet  
Search completed
```

**Pending manual review****Submitted**